

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO 5: Construct interoperable web services by leveraging JAX-RPC, WSDL, SOAP, and XML Schema for seamless data exchange across distributed systems. (L4, L5)

Assessment Tool: Viva Voce | Weightage: 10%

Student Name: K Phani Sridhar Yogi

Register No.: 192311091

VIVA VOCE – QUESTIONS AND ANSWERS

1. Define SOAP.

SOAP (Simple Object Access Protocol) is an XML-based messaging protocol used to exchange structured information between applications over a network. It supports standards such as headers, security, transactions, and fault handling and is commonly transported over HTTP or HTTPS.

2. Define WSDL.

WSDL (Web Services Description Language) is an XML-based language used to describe a web service. It specifies the operations offered by the service, the messages and data types involved, the communication binding, and the service endpoint.

3. What is an API?

An API (Application Programming Interface) is a defined set of rules and endpoints that allows one software application to communicate with another. For example, a mobile application can use an API to request customer or product data from a server.

4. Define XML Schema.

XML Schema, commonly called XSD, defines the structure and data types of an XML document. It specifies elements, attributes, data types, occurrence rules, and relationships so that XML messages can be validated.

5. What is a Web Service?

A web service is a software component accessible over a network that allows different applications to exchange data or functionality using standard web technologies. SOAP and REST are common approaches for implementing web services.

6. Define REST.

REST (Representational State Transfer) is an architectural style for designing networked applications around resources. REST commonly uses HTTP methods such as GET, POST, PUT, PATCH, and DELETE and often exchanges data using JSON.

7. What is a Protocol?

A protocol is a set of rules that defines how data is formatted, transmitted, received, and interpreted between communicating systems. HTTP, HTTPS, TCP, and SOAP are examples of technologies or standards used in web communication.

8. Define Endpoint in web services.

An endpoint is the specific network address or URL through which a client accesses a web service or API operation. For example, an endpoint might be `https://example.com/api/patients`.

9. What is a Service in web applications?

A service is a software component that provides a specific function or business capability to other applications through a defined interface. Examples include authentication, payment, patient-management, and notification services.

10. What is the difference between SOAP and REST?

SOAP is a protocol based on structured XML messages and supports enterprise standards such as WS-Security. REST is an architectural style that uses standard HTTP operations and commonly uses JSON. REST is generally simpler and lighter, while SOAP is useful when formal contracts, message-level security, or enterprise WS-* features are required.

11. What is the role of WSDL in web services?

WSDL acts as a formal service contract. It tells clients what operations are available, what input and output messages look like, what data types are used, which communication binding is supported, and where the service can be accessed.

12. What is JSON and how is it used in APIs?

JSON (JavaScript Object Notation) is a lightweight text-based data format consisting of objects, arrays, key-value pairs, strings, numbers, booleans, and null values. REST APIs commonly use JSON to exchange data because it is compact and easy for web and mobile applications to parse.

13. What is HTTP and why is it used in web services?

HTTP (Hypertext Transfer Protocol) is an application-layer protocol used to transfer requests and responses between clients and servers. Web services use HTTP because it is widely supported and provides standard methods, status codes, headers, and communication semantics.

14. What is meant by request and response in web communication?

A request is a message sent by a client to a server asking for an operation or resource. A response is the server's message containing the result, data, status code, or an error. For example, a client can send a GET request and receive a JSON response.

15. What is the importance of XML in web services?

XML provides a structured, platform-independent format for representing and exchanging data. It is especially important in SOAP web services because SOAP messages use XML and can be validated using XML Schema.

16. What is the purpose of an API endpoint?

An API endpoint provides a specific address through which a client can access a particular resource or operation. It defines where the request should be sent and, together with the HTTP method, determines what action is performed.

17. What is the difference between HTTP and HTTPS?

HTTP sends application data without transport-layer encryption. HTTPS is HTTP protected by TLS, which provides encryption, server authentication through certificates, and protection against tampering during transmission. HTTPS is essential when sensitive information is exchanged.

18. What is meant by RESTful services?

RESTful services are web services designed according to REST principles. They use resource-based URLs, standard HTTP methods, stateless communication, appropriate status codes, and commonly JSON representations to exchange resource data.

19. What is a namespace in XML?

An XML namespace uniquely identifies element and attribute names so that names from different XML vocabularies do not conflict. A namespace is usually identified by a URI and declared using xmlns.

20. What is the role of a client and server in web services?

The client initiates a request for a resource or operation, while the server receives the request, processes it, and returns a response. For example, a mobile app acts as the client and a web-service application acts as the server.

QUICK VIVA REVISION POINTS

For quick preparation, remember the following key relationships:

- SOAP → XML-based messaging protocol.
- WSDL → Describes the service contract and interface.
- XSD → Defines and validates XML data structure.
- API → Interface that enables software-to-software communication.
- REST → Architectural style using resources and HTTP.
- HTTP/HTTPS → Common transport/application communication mechanisms for web services.
- JSON → Lightweight data format commonly used by REST APIs.
- Endpoint → URL/address used to access an API or web-service operation.
- Client → Sends requests; Server → Processes requests and returns responses.

CODE OF CONDUCT

I, M.PAVITHRAN / 192411014 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: PAVITHRAN

Name:PAVITHRAN

Register No.: 192411014

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 / Level 1
Conceptual Understanding	25	Deep understanding with complete clarity	Correct explanation with minor gaps	Partial understanding / unable to explain basics
Application	25	Effectively applies concepts to new situations	Applies concepts with minor errors	Limited or ineffective application
Analytical Thinking	20	Critical analysis with strong justification	Reasonable analysis with some justification	Superficial or no analytical reasoning
Communication	15	Clear, confident and appropriate terminology	Clear with minor hesitation	Unclear or ineffective communication
Response to Questions	15	Accurate, confident, detailed responses	Correct but limited depth	Partial or unable to respond